Geometric Median

A study on a variation of Fermat's point in different types of triangles

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Introduction

The Fermat point (or Torricelli point) in a triangle is the point which minimises the sum of distances to each vertex and has many applications in real life such as in minimising distances in networks. This project aims to explore a variant of the geometric median in triangles, which is when the product is maximised whilst originally the sum is minimised. Such a point of max product may have special properties that can be used in real life, for example in applications like finding structural weakness.

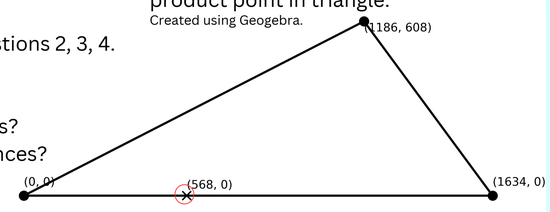
Research Questions

For each research question, the point is to be found and a formula for its value created.

Research Question 1 is included as a precursor to our main Research Questions 2, 3, 4.

1. What point in a triangle minimises the sum of distances?
2. What point in an equilateral triangle maximises the product of distances?
3. What point in an isosceles triangle that maximises the product of distances?
4. What point in a general triangle maximises the product of distances?

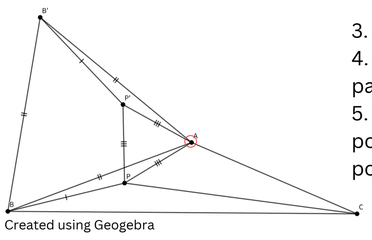
Diagram showing approximated max product point in triangle.
Created using Geogebra.



RQ1: Fermat's point

There are 2 cases, when the triangle is obtuse and when it is not.

Case 1: There exists an angle ≥ 120° in △ABC.
Claim: The obtuse vertex is the Fermat point
1. Consider any point p in the interior of the triangle
2. Rotate △ABP 60° clockwise about A. Let the image of point B be B' and point P be P' after rotation, such that B' and C are on opposite sides of the line AB.



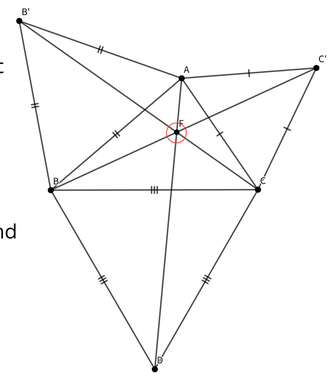
Created using Geogebra

3. △ABP congruent to △AB'P', APP' equilateral, ABB' equilateral.
4. |CP|+|AP|+|BP| = |CP| + |PP'| + |P'B'| which is the length of the path CPP'B'
5. The length of the path CPP'B' is minimised when point P' = point P = point A, i.e. point A, the obtuse angle, is the Fermat point.

Case 2: All angles are < 120° in △ABC
Claim: The fermat point can be found as follows: Construct three equilateral triangles out of the three sides from △ABC, then connect each new vertex to the respective opposite vertex. The added lines will concur at Fermat's point.
1. We use a similar method as used in Case 1 to prove the sum of distances is minimal at F
2. Notice that △ACB' is congruent to △AC'B by rotation.
Thus ∠ABC' = ∠AB'C and ∠AC'B = ∠ACB', hence AB'BF concyclic and AC'CF concyclic. Hence F is a unique point.

The exact minimum value of sum of distances in case 2 is found by cosine rule and Heron's formula:

$$\sqrt{\frac{1}{2}\left(a^2+b^2+c^2+\sqrt{3(a+b+c)(-a+b+c)(a-b+c)(a+b-c)}\right)}$$



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RQ2: Max Product in Equilateral Triangle

We first prove that it is always optimal for the point to lie on the boundary of the triangle. Then we prove that if the max point lies on the boundary, it is the midpoint of an edge. Let the vertices be A(0, 0), B(2, 0), C(1, $\sqrt{3}$)

Claim 1: Point on boundary will yield greater product of distances than point within boundary
1. Consider a variable point P in the interior of the triangle. WLOG P lies in the shaded region, by similar triangles:

$$h \le \frac{2}{\sqrt{3}} \qquad c \le |FD| \le \frac{h}{\sqrt{3}}$$

2. Construct line ED through P such that ED is parallel to AB.
3. Set up product expression of point P, in terms of distance from midpoint F of ED.

$$|AP|\cdot|BP|\cdot|CP| = \sqrt{(1+c)^2+\left(\sqrt{3}-h\right)^2}\sqrt{(1-c)^2+\left(\sqrt{3}-h\right)^2}\sqrt{h^2+c^2}$$

4. Differentiate product expression and do second derivative test.

$$3c^4+(8-8\sqrt{3}h+6h^2)c^2+\left(16-16\sqrt{3}h+24h^2-8\sqrt{3}h^3+3h^4\right)=0$$

5. It is obtained that midpoint F of line ED is a minimum point, i.e. when $c=0$
6. Hence maximum value will be attained when point P is shifted along ED onto the boundary of the triangle

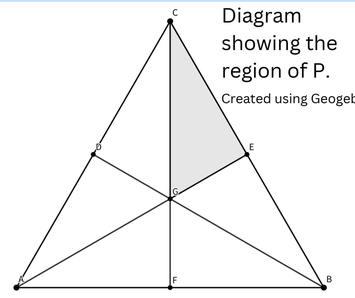


Diagram showing the region of P.
Created using Geogebra

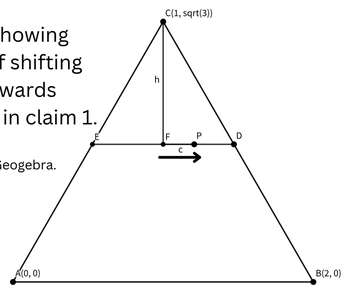


Diagram showing process of shifting point P towards boundary in claim 1.
Created using Geogebra.

Claim 2: Max point lies on the midpoint of the edge
1. Set up product expression of point E, in terms of distance from midpoint M of a side.

$$\left(\sqrt{3+k^2}\right)(1-k)(1+k)$$

2. Smooth the product expression towards the midpoint

$$\sqrt{\left(3-2k^2-k^4\right)\left(1-k^2\right)} \le \sqrt{3}$$

3. It is obtained that the product expression increases towards midpoint, hence, the point of max product in equilateral triangle is obtained at the midpoint of the sides

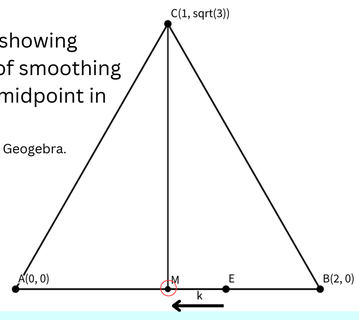


Diagram showing process of smoothing towards midpoint in claim 2.
Created using Geogebra.

RQ3: Max Product in Isosceles Triangle

We could not carry out the same proof as we did in RQ2 to prove that the max point lies on the boundary of an isosceles triangle. We quote Binyamin Schwarz in 1936 [1] and Gulbank D. Chakerian with Murray S. Klamkin in 2007 [2], who show that the point which maximises the product of distances within any n-gon always lies on the boundary of the n-gon.

We first calculate the max point on the base of the isosceles triangle, then calculate the max on either side, then compare both values. Let the vertices of the isosceles be A(0, 0), B(2, 0), C(1, t)

Claim 1:
When height of the isosceles triangle $t > \frac{1}{\sqrt{2}}$, the max value on base is t

Else, the max value on base is $= \frac{2\sqrt{(1-2t^2)^3}}{3\sqrt{3}}$

1. Product of distances $= \left((1-k)(1+k)\sqrt{k^2+t^2}\right)$
2. Differentiate this expression and equate it to 0 to find the stationary points. We get the following result: $k=0$ or $k=\frac{\sqrt{1-2t^2}}{\sqrt{3}}$
3. By using the second derivative test, we manage to obtain which value of k is the maximum point for different ranges of t

Claim 2:
The max point of isosceles triangle lies on the longest side of the triangle.

1. We firstly assume that the max point is found on the leg of the triangle and use a similar method as shown in Claim 1 to calculate the location of the max point
2. We then compare the maximum attainable product of distances between if the max point was assumed to be on the base VS if it was assumed to be on the leg to find where the max point is for a specific range of values
3. By plotting a graph as shown, we obtain the result that max point is always found on the longest side of the triangle

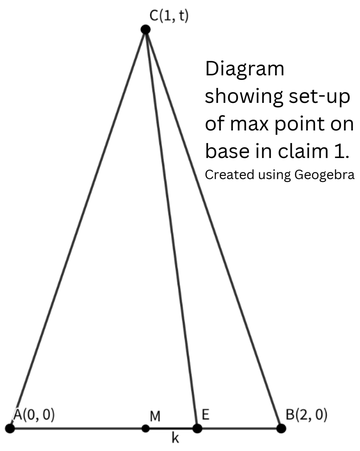
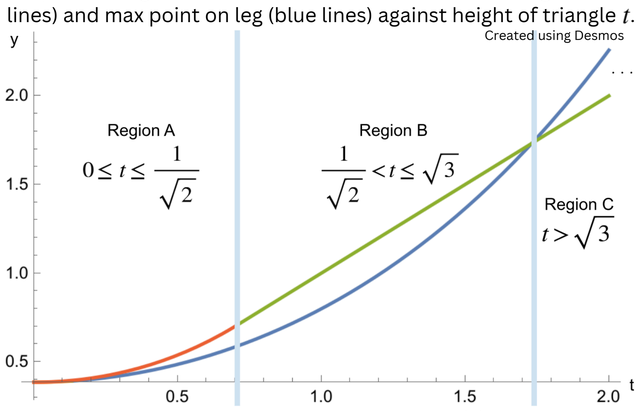


Diagram showing set-up of max point on base in claim 1.
Created using Geogebra

Graph showing value of the max point on base (red and green lines) and max point on leg (blue lines) against height of triangle t.
Created using Desmos



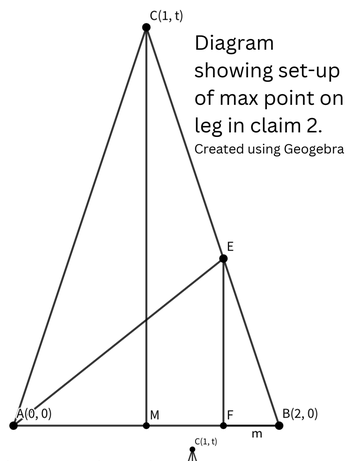


Diagram showing set-up of max point on leg in claim 2.
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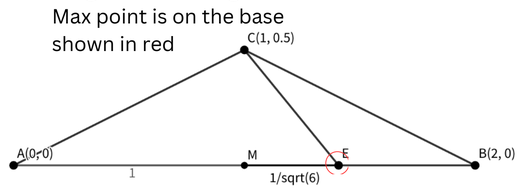
Region A triangle

Max point is on the base shown in red



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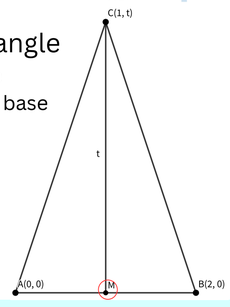
Region B triangle

Max point is the midpoint of the base



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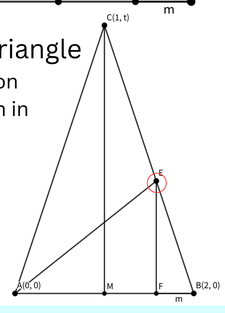
Region C triangle

Max point is on the leg shown in red

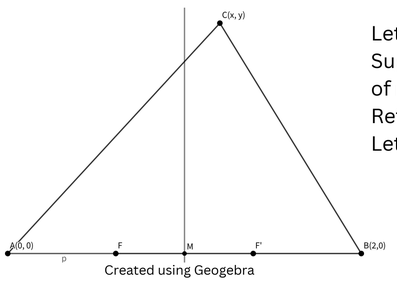


Created using Geogebra

RQ4: Max Product in General Triangle

We calculate the max value on each side, then we compare the max values to prove that the max point always lies on the longest side.

We calculate the max values on each side as follows:
Let the the 3 vertices be A(0, 0), B(2, 0), C(x, y), Without loss of generality, |BA| ≥ |AC| ≥ |CB|.
1. Bound the location of the max point by reflection across midpoint



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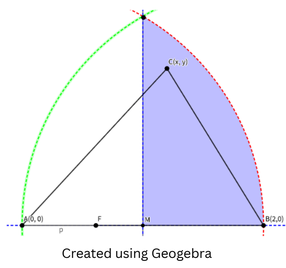
Let p be the distance |BF'|
Suppose that F' is the max point along AB that lies on the right of midpoint M.
Reflect F' across the perpendicular bisector to obtain point F.
Let |BF'|=|AF|=p

Claim: Product at point F is always greater than product at point F'.

$|AF|\cdot|BF|\cdot|CF| > |AF'|\cdot|BF'|\cdot|CF'|$
$\Leftrightarrow (p)(2-p)|CF| > (2-p)(p)|CF'|$
$\Leftrightarrow |CF| > |CF'|$
$\Leftrightarrow \sqrt{(x-p)^2+y^2} > \sqrt{(2-x-p)^2+y^2}$
$\Leftrightarrow (x-p)^2 > (2-x-p)^2$
$\Leftrightarrow x+p > 1$ this is true by setup



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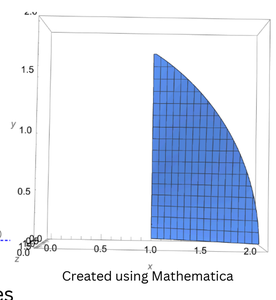


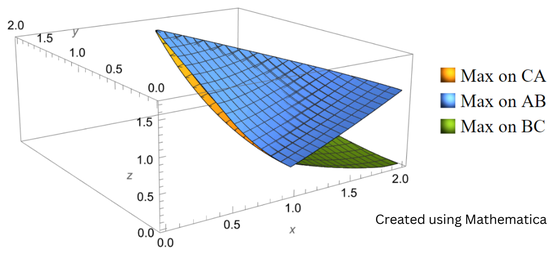
Created using Mathematica

2. Differentiate for all three sides
3. Solve derivative = 0 for all three sides. We obtain this equation:

$$-3p^3+p^2(4+5x)-2p(3x+x^2+y^2)+2(x^2+y^2)=0$$

The (x, y) of the max point is the solution of this cubic equation.

We use a 3D graph to compare the maximum values of the product of distances when the point lies on AB, BC and CA.



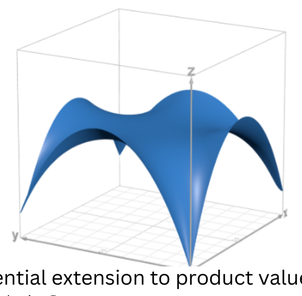
Created using Mathematica

It can be observed that the blue plane which represents the max on side AB, the longest side, is always on top, i.e. max point lies on the longest side.

Applications and Extensions

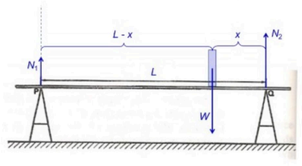
Applications of the max product point in a triangle may include locating the weakest point of a 2-D object
- Weakest point of a beam is the midpoint (proven with calculus)
- Used to identify structural flaws in certain designs

Some potential extensions of our project include
- Locating the point in general n-gons (square is shown on the right)
- Finding the weakest point in a shape made of a 'weighted' material
- Exploring the concept of a 'weakest plane' of a 3D object
- Conduct analysis on appropriate formulas



Potential extension to product value in square
created using Desmos

Image sourced from HCI O-level physics notes



$$\sigma_b = \frac{My}{I}$$

σ_b – Bending stress
M – Calculated bending moment
y – Vertical distance away from the neutral axis
I – Moment of inertia around the neutral axis

References

[1] Schwarz, B. (1965). On the product of the distances of a point from the vertices of a polytope. Israel Journal of Mathematics, 3(1), 29–38: 10.1007/BF02760025. Sci. (n.d.-a). https://link.springer.com/article/10.1007/BF02760025

[2] Klamkin, S., Chakerian, D. (2007). The product of the distances of a point inside a regular polytope to its vertices. Journal of Geometry, 86(1-2), 31 41: 10.1007/S00022-006-1813-7. Sci. (n.d.). https://link.springer.com/article/10.1007/s00022-006-1813-7